

1

2

3

4

A

A

B

B

C

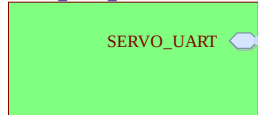
C

D

D

Connectors for servos. Driver:
converts full - duplex UART
into half - duplex UART to
communicate with servos with
only one data - line.

U_St3025_Servo_Schematic
St3025_Servo_Schematic.SchDoc



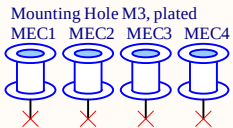
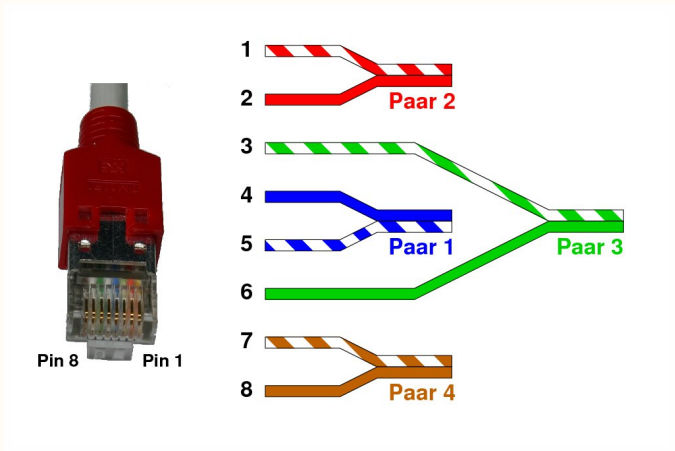
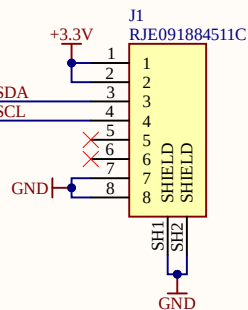
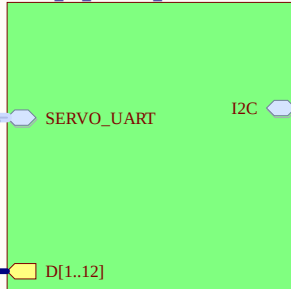
U_LED_16x_Schematic
LED_16x_Schematic.SchDoc



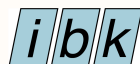
12 4x4 WS2812B - matrices.
Represent the seconds.

ESP32S3 Wroom 2
Schematic with USB - C,
UART - bridge, LDO and
PowerJack.

U_ESP32_S3_Wroom2_Schematic
ESP32_S3_Wroom2_Schematic.SchDoc



ETH
Eidgenössische Technische Hochschule Zürich
Swiss Federal Institute of Technology Zurich



Date: 03.08.2026 Time: 10:44:42
Revision: V0.1

Drawn by: Manuel Hueppi
Institut: IBK
Departement: DBAUG

Project: BauhallenUhr.PrjPcb
Page: 1 from: 4

File: C:\Users\hueppim\Documents\Github\BauhallenUhr\Altium_BauhallenUhr_V0.1\Blackbox_BauhallenUhr_Schematic.SchDoc

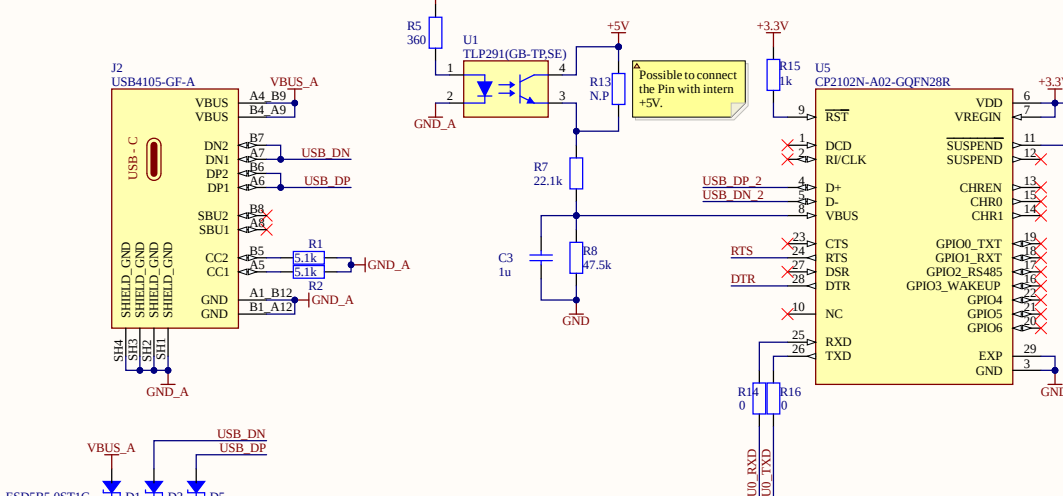
1

2

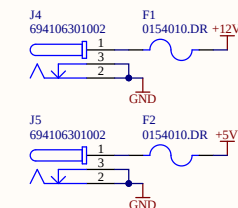
3

4

USB to UART



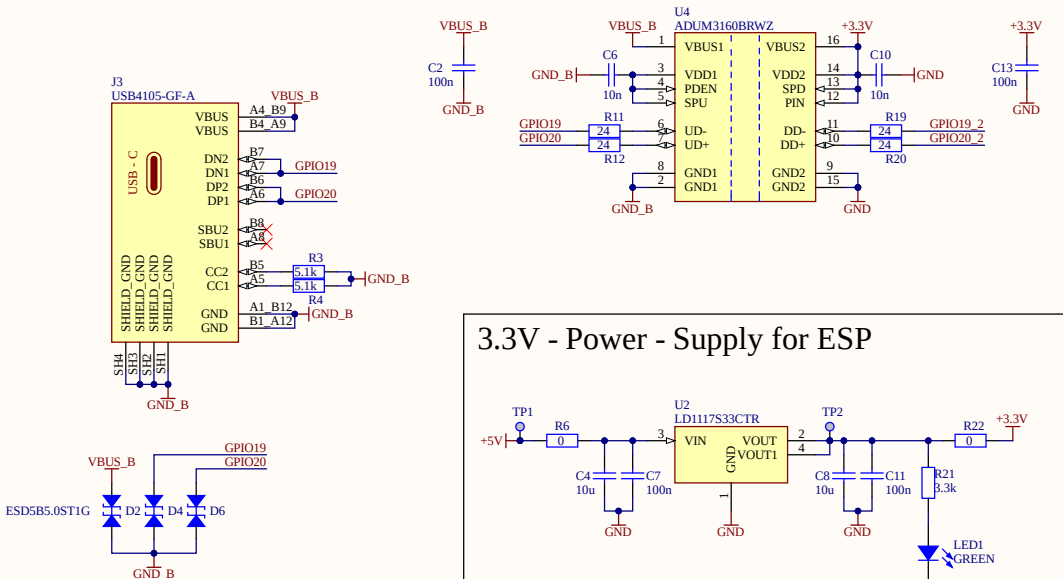
Power - Supply



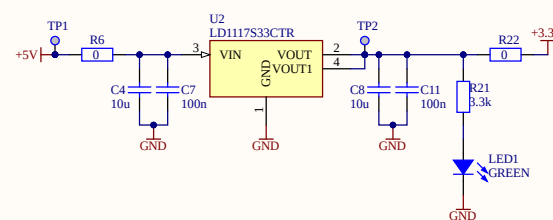
I2C Connector Entries



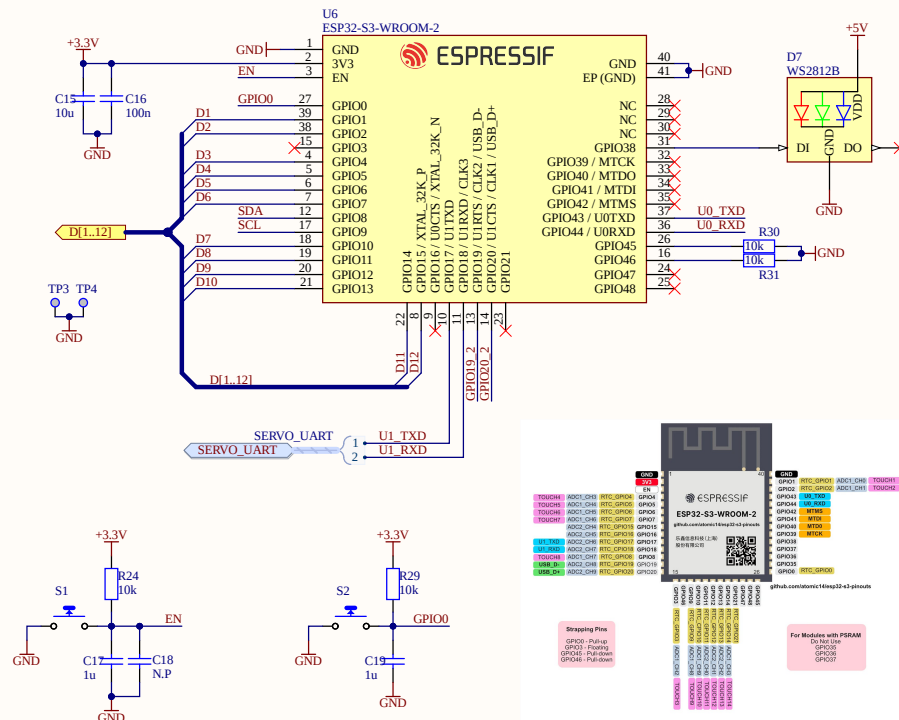
ESP USB



3.3V - Power - Supply for ESP



ESP32S3 Wroom 2 Module



ETH

Eidgenössische Technische Hochschule Zürich
Swiss Federal Institute of Technology Zurich



Title of drawing: *ESP32_S3_Wroom2_Schematic.SchDoc*

Date: 03.08.2026
Revision: V0.1

Time: 10:44:42

Drawn by: Manuel Hueppi
Institut: IBK
Departement: DBAUG

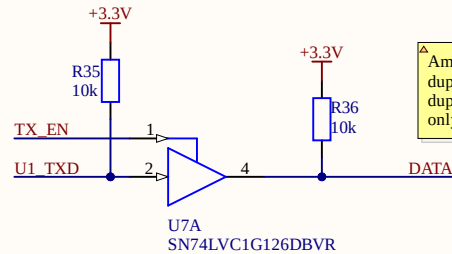
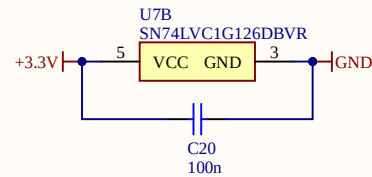
Project: *BauhallenUhr.PrjPcb*
Page: 2 from: 4

File: C:\Users\hueppim\Documents\Github\BauhallenUhr\Altium_BauhallenUhr_V0.1\ESP32_S3_Wroom2_Schematic.SchDoc

UART Connectors

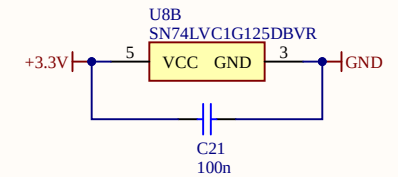


Single Bus Buffer Gate



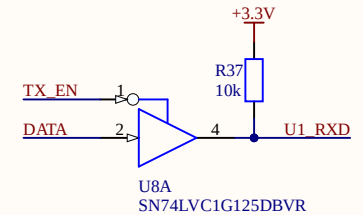
Amplifier to convert full - duplex UART to half - duplex, because the servos only have one data - line.

Inverted Bus Buffer Gate

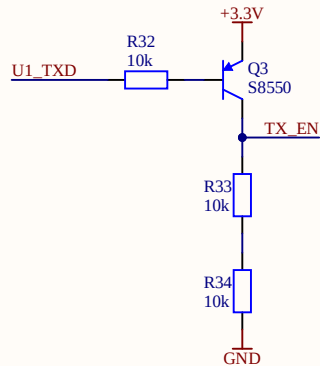


Find replacement for SN74LVC1G125DBVR!!!!

Amplifier to convert full - duplex UART to half - duplex, because the servos only have one data - line.



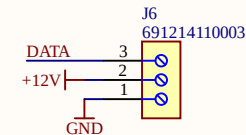
Half - Duplex Direction Detection



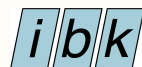
If the ESP32 writes something to the servos, the TX_EN - signal will enable the data flow between the SERVO_TXD - line and the DATA - line. If the ESP32 writes nothing, the data flow in the opposite direction is enabled, which means the ESP32 can read e. g the position of the servos.

Servo Connector

Only one connector, because these servos are controlled via UART, they can be chained together (serial).



ETH
Eidgenössische Technische Hochschule Zürich
Swiss Federal Institute of Technology Zurich



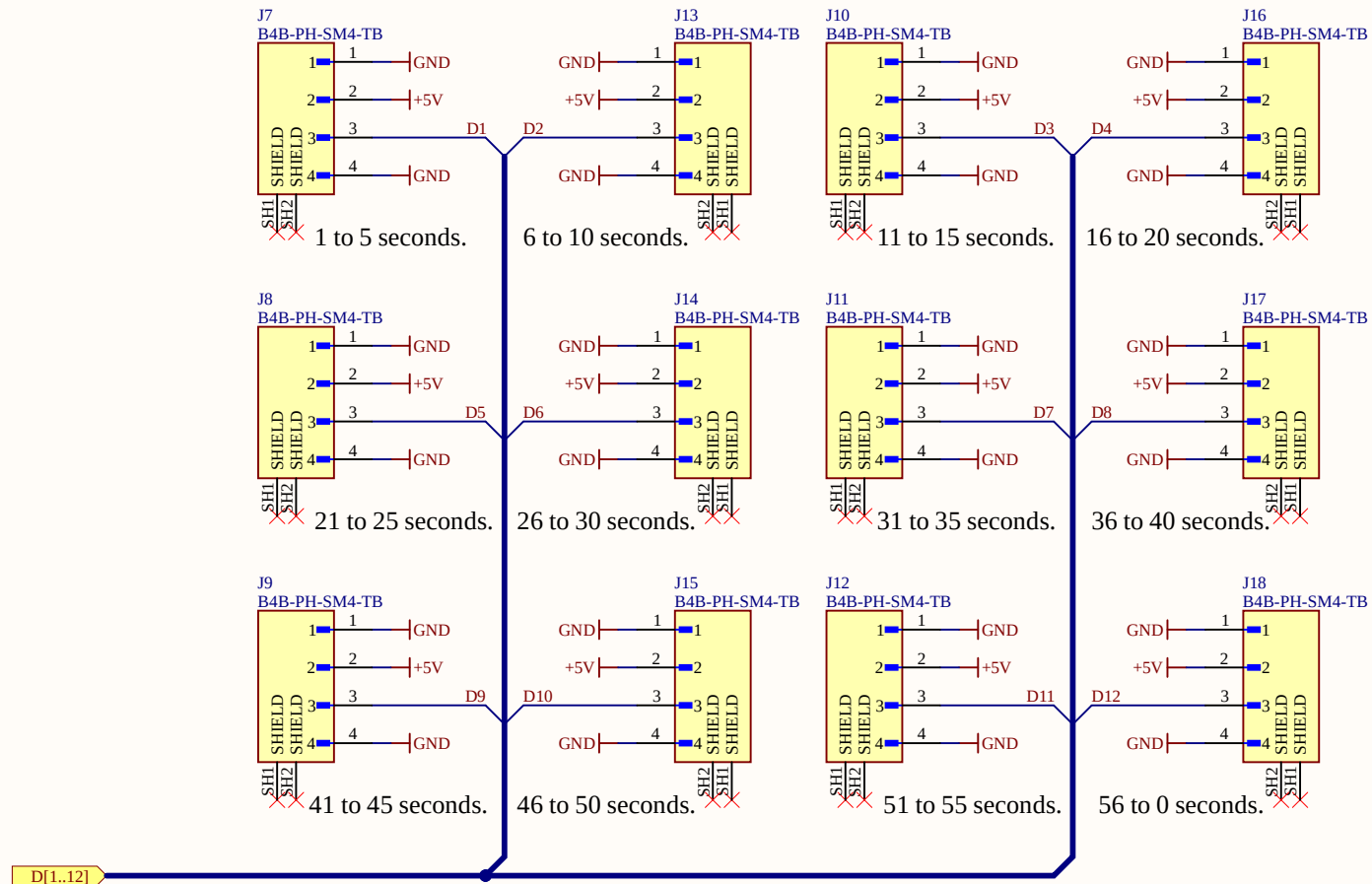
Title of drawing: St3025_Servo_Schematic.SchDoc

Date: 03.08.2026 Time: 10:44:42
Revision: V0.1

Drawn by: Manuel Hueppi
Institut: IBK
Departement: DBAUG

Project: BauhallenUhr.PrjPcb
Page: 3 from: 4

File: C:\Users\hueppim\Documents\Github\BauhallenUhr\Altium_BauhallenUhr_V0.1\St3025_Servo_Schematic.SchDoc



Board Stack Report